Submission Summary

Conference Name

3rd International Conference on Innovative Trends in Electrical, Electronics and Bio-Technology Engineering

Track Name

Track 2: Artificial Intelligence, Machine Learning, Image Processing

Paper ID

268

Paper Title

Implementation Of an Intelligent Vision Transformer-Based Deep Learning Model for Feedstock Quality Assessment and Defect Detection in Industry 4.0 Production

Abstract

The rapid adoption of Industry 4.0 technologies has increased the demand for intelligent and automated quality inspection systems in modern manufacturing environments. Feedstock quality plays a critical role in determining the performance, reliability, and efficiency of industrial production processes. Traditional inspection methods often rely on manual evaluation, which can be time-consuming, inconsistent, and prone to human error. To address these challenges, this paper presents the implementation of an intelligent Vision Transformer (ViT)-based deep learning model for automated feedstock quality assessment and defect detection. The proposed system leverages the self-attention mechanism of Vision Transformers to extract complex visual features from feedstock images and accurately identify quality variations and surface defects. Image preprocessing and feature extraction techniques are employed to enhance data representation and improve model performance. The model is trained and evaluated on a labeled feedstock image dataset, enabling robust classification of defective and non-defective samples. By integrating advanced deep learning with industrial inspection workflows, the proposed approach supports real-time decision-making and improves production quality control. Experimental analysis demonstrates that the Vision Transformer architecture provides effective defect detection capability with improved accuracy and scalability compared to conventional inspection approaches. The proposed framework contributes to the advancement of smart manufacturing by enabling automated, reliable, and intelligent feedstock quality monitoring within Industry 4.0 production systems.

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Primary Subject Area

Pattern Analysis & Image Processing

Secondary Subject Areas

Machine Vision and Machine Learning

Submission Files

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